The Exact $4+4+4+1$ Continuation: A Thirteen-Dimensional Quaternionic Peirce Slice, Its Dirac–Lorentz Cubic, and Its Rank-Defect S^{12}

Working theorem/proof module

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Abstract

The previously established marked observation matrix

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

splits $\mathrm{Herm}_3(\mathbb{C})$ as a four-dimensional Lorentz vector, one four-real-dimensional Weyl spinor, and a scalar. This note extends the same map over the quaternions without replacing its basis or its coefficients. Fixing the observed block to remain in $\mathrm{Herm}_2(\mathbb{C})$ gives the pullback

$$\mathcal{J}_{13,P} := (\mathcal{O}_P^{\mathbb{H}})^{-1}(\mathrm{Herm}_2(\mathbb{C})) \subset \mathrm{Herm}_3(\mathbb{H}),$$

a real thirteen-dimensional space with the exact decomposition

$$\mathcal{J}_{13,P} \cong \mathrm{Herm}_2(\mathbb{C}) \oplus \mathbb{C}^2 \oplus \mathbb{C}^2 \oplus \mathbb{R}.$$

Its restricted Moore determinant is the cubic

$$\mathcal{N}_{13}(G, z, w, \lambda) = \lambda \det G - z^* \operatorname{adj}(G) z - w^* \operatorname{adj}(G) w.$$

After charge conjugating the second spinor this becomes the Lorentz-invariant vector–Dirac-spinor–scalar expression

$$\lambda \det G - \psi_L^* \operatorname{adj}(G) \psi_L - \psi_R^* G \psi_R.$$

We classify every positive-semidefinite fibre, prove invariance under $(SL_2(\mathbb{C}) \times SU(2))/\{\pm(1,1)\}$, identify an $\mathbb{CP}^1$-family of determinant-preserving nine-dimensional slices, and show that the old compactified rank defect is the fixed unknotted $S^8 \subset S^{12}$. The complement is $S^3 \times \mathbb{R}^9$, so the added four-dimensional spinor has the unit-quaternion linking sphere predicted by the coordinate count. The slice is not a Jordan subalgebra; an explicit product leaves it, and its Jordan closure is all of $\mathrm{Herm}_3(\mathbb{H})$. We also compare it, on the identical coordinate space, with the other natural thirteen-dimensional complex 4×4 determinant model and prove the exact correction

$$\det \begin{pmatrix} G & Z \\ Z^* & \lambda I_2 \end{pmatrix} = \lambda \mathcal{N}_{13}(G, Z, \lambda) + |\det Z|^2.$$

Thus the quaternionic model is distinguished by preserving the old cubic exactly, while the complex rank-four model carries an additional Pluecker term. No toroidal filling or quantum field theory is identified by the dimension count alone.

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1 The retained observation map and the quaternionic pullback

Write the quaternions as

$$\mathbb{H} = \mathbb{C} \oplus \mathbb{C}\mathbf{j}, \qquad \mathbf{j}^2 = -1, \qquad \mathbf{j}a = \overline{a}\,\mathbf{j} \quad (a \in \mathbb{C}). \tag{1}$$

For $q = a + b\mathbf{j}$, quaternionic conjugation is $\overline{q} = \overline{a} - b\mathbf{j}$.

Retain the exact marked data

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad C = PP^* = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \tag{2}$$

$$R_P = P^* C^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix}, \qquad \Pi_u = \frac{uu^*}{3}, \qquad X = (R_P\ u). \tag{3}$$

Then

$$PR_P = I_2, \quad R_P P = I_3 - \Pi_u, \quad u^* R_P = 0, \quad Pu = 0, \quad \det X = 1, \quad X^{-1} = \begin{pmatrix} P \\ u^*/3 \end{pmatrix}. \tag{4}$$

Extend the observation map, using the same real matrix P, to

$$\mathcal{O}_P^{\mathbb{H}} : \mathrm{Herm}_3(\mathbb{H}) \longrightarrow \mathrm{Herm}_2(\mathbb{H}), \qquad \mathcal{O}_P^{\mathbb{H}}(H) = PHP^*. \tag{5}$$

Fix the displayed complex subalgebra $\mathbb{C} \subset \mathbb{H}$ and define

$$\boxed{\mathcal{J}_{13,P} := (\mathcal{O}_P^{\mathbb{H}})^{-1}(\mathrm{Herm}_2(\mathbb{C})).} \tag{6}$$

Thus $\mathcal{J}_{13,P}$ is defined by a pullback square, not by dimension alone:

$$\begin{array}{ccc} \mathcal{J}_{13,P} & \longrightarrow & \mathrm{Herm}_3(\mathbb{H}) \\ {\scriptstyle \mathcal{O}_P}\big\downarrow & & \big\downarrow{\scriptstyle \mathcal{O}_P^{\mathbb{H}}} \\ \mathrm{Herm}_2(\mathbb{C}) & \longrightarrow & \mathrm{Herm}_2(\mathbb{H}). \end{array} \tag{7}$$

Theorem 1.1 (Unique $4+4+4+1$ coordinates)**.** *Every $H \in \mathcal{J}_{13,P}$ has one and only one expression*

$$H = X \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} X^*, \qquad G \in \mathrm{Herm}_2(\mathbb{C}), \quad \psi \in \mathbb{H}^2, \quad \lambda \in \mathbb{R}. \tag{8}$$

Writing

$$\psi = z + w\mathbf{j}, \qquad z, w \in \mathbb{C}^2, \tag{9}$$

the extraction formulae are

$$G = PHP^*, \qquad z + w\mathbf{j} = \frac{1}{3}PHu, \qquad \lambda = \frac{1}{9}u^*Hu. \tag{10}$$

Consequently

$$\boxed{\mathcal{J}_{13,P} \cong_{\mathbb{R}} \underbrace{\mathrm{Herm}_2(\mathbb{C})}_{4} \oplus \underbrace{\mathbb{C}^2}_{4} \oplus \underbrace{\mathbb{C}^2}_{4} \oplus \underbrace{\mathbb{R}}_{1}}. \tag{11}$$

The old space $\mathrm{Herm}_3(\mathbb{C})$ *is exactly the subspace $w = 0$.*

Proof. For an arbitrary quaternionic Hermitian matrix, conjugation by X^{-1} gives

$$X^{-1}HX^{-*} = \begin{pmatrix} PHP^* & PHu/3 \\ u^*HP^*/3 & u^*Hu/9 \end{pmatrix}.$$

Membership in $\mathcal{J}_{13,P}$ says precisely that the upper-left block is complex Hermitian. The remaining off-diagonal block is an arbitrary vector in $\mathbb{H}^2$, and (1) gives its unique pair (z, w). The dimension is $4 + 8 + 1 = 13$. If $w = 0$, every block is complex, and because X is real this is exactly $\mathrm{Herm}_3(\mathbb{C})$; conversely every complex Hermitian matrix has $w = 0$. □

2 The restricted Moore determinant

Use the complex 2×2 representation

$$\rho(a + b\mathbf{j}) = \begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix}. \tag{12}$$

For a quaternionic Hermitian matrix M, its Moore determinant $\det_{\mathbb{H}} M$ is the real polynomial normalized by $\det_{\mathbb{H}} I = 1$ and

$$\det_{\mathbb{C}} \rho(M) = (\det_{\mathbb{H}} M)^2. \tag{13}$$

Lemma 2.1 (Quaternionic form splitting)**.** *If $A \in \mathrm{Herm}_2(\mathbb{C})$ and $\psi = z + w\mathbf{j} \in \mathbb{H}^2$, then*

$$\boxed{\psi^* A \psi = z^* A z + w^* A w \in \mathbb{R}.} \tag{14}$$

Proof. Expand with $\mathbf{j}a = \overline{a}\mathbf{j}$. The two complex scalar parts are z^*Az and w^*Aw. The coefficient of $\mathbf{j}$ is $z^*Aw - w^{\mathsf{T}}\overline{A}\,\overline{z}$, which vanishes because $A = A^*$. Both scalar terms are real. □

Theorem 2.2 (The thirteen-dimensional cubic)**.** *For H with coordinates (G, z, w, λ), the restricted Moore determinant is*

$$\boxed{\mathcal{N}_{13}(G, z, w, \lambda) := \det_{\mathbb{H}} H = \lambda \det_{\mathbb{C}} G - z^* \operatorname{adj}(G) z - w^* \operatorname{adj}(G) w.} \tag{15}$$

Equivalently, with $Z = (z\ w) \in \mathrm{Mat}_{2\times 2}(\mathbb{C})$,

$$\mathcal{N}_{13}(G, Z, \lambda) = \lambda \det G - \operatorname{tr}(Z^* \operatorname{adj}(G) Z). \tag{16}$$

On $w = 0$ this is exactly the old nine-dimensional cubic

$$\mathcal{N}_9(G, z, \lambda) = \lambda \det G - z^* \operatorname{adj}(G) z, \tag{17}$$

with no extra factor.

Proof. Since $\det X = 1$, quaternionic congruence by X leaves the Moore determinant unchanged. For invertible G, the quaternionic Schur formula gives

$$\det_{\mathbb{H}} \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} = \det G\,(\lambda - \psi^* G^{-1} \psi) = \lambda \det G - \psi^* \operatorname{adj}(G) \psi.$$

Both sides are polynomial, so the identity holds for singular G as well. Apply Theorem 2.1 with $A = \operatorname{adj}(G)$. □

2.1 Dirac form of the same cubic

Put

$$\epsilon_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{18}$$

and write the second left-handed spinor as the charge conjugate of a right-handed one:

$$w = \epsilon_2 \overline{\psi_R}, \qquad z = \psi_L. \tag{19}$$

For every $G \in \mathrm{Herm}_2(\mathbb{C})$,

$$\epsilon_2^* \operatorname{adj}(G) \epsilon_2 = G^{\mathsf{T}}. \tag{20}$$

Corollary 2.3 (Vector–Dirac-spinor–scalar cubic)**.** *In the variables $(G, \psi_L, \psi_R, \lambda)$,*

$$\boxed{\mathcal{N}_{13} = \lambda \det G - \psi_L^* \operatorname{adj}(G) \psi_L - \psi_R^* G \psi_R.} \tag{21}$$

Under $g \in SL_2(\mathbb{C})$,

$$G \mapsto g G g^*, \qquad \psi_L \mapsto g \psi_L, \qquad \psi_R \mapsto (g^*)^{-1} \psi_R, \qquad \lambda \mapsto \lambda. \tag{22}$$

Thus the two four-real-dimensional summands may be read either as two left Weyl spinors or, after the exact map (19)*, as one Dirac spinor.*

Proof. Equation (20) gives

$$w^* \operatorname{adj}(G) w = \psi_R^{\mathsf{T}} G^{\mathsf{T}} \overline{\psi_R} = \psi_R^* G \psi_R.$$

The identity $g\epsilon_2 = \epsilon_2 (g^{\mathsf{T}})^{-1}$ proves that $w \mapsto gw$ is equivalent to the right-handed transformation in (22). □

3 The complete positive cone and all observation fibres

Let

$$\mathcal{C}_{13,P} = \mathcal{J}_{13,P} \cap \mathrm{Herm}_3(\mathbb{H})_{\succeq 0}. \tag{23}$$

Theorem 3.1 (Exact positivity and rank)**.** *A point (G, z, w, λ) lies in $\mathcal{C}_{13,P}$ if and only if*

$$\boxed{G \succeq 0, \qquad z, w \in \operatorname{Ran} G, \qquad \lambda \geq z^* G^\dagger z + w^* G^\dagger w.} \tag{24}$$

Put

$$s = \lambda - z^* G^\dagger z - w^* G^\dagger w \geq 0. \tag{25}$$

Then the quaternionic rank is

$$\boxed{\operatorname{rank}_{\mathbb{H}} H = \operatorname{rank}_{\mathbb{C}} G + \begin{cases} 1, & s > 0, \\ 0, & s = 0. \end{cases}} \tag{26}$$

For fixed $G \succeq 0$ of rank r, the positive fibre of $\mathcal{O}_P$ is

$$\{(z, w, s) : z, w \in \operatorname{Ran} G,\ s \geq 0\} \cong \mathbb{C}^{2r} \times [0, \infty), \tag{27}$$

of real dimension $4r + 1$. In particular,

$$\boxed{\mathcal{O}_P^{-1}(0) \cap \mathcal{C}_{13,P} = \{\lambda u u^* : \lambda \geq 0\}.} \tag{28}$$

Proof. Work in the block basis X. A positive block matrix must have $G \succeq 0$. Positivity on vectors from $\ker G \oplus \mathbb{H}$ forces $\psi = z + w\mathbf{j}$ to lie in the quaternionic range of G. Since G is complex, this is equivalent to $z, w \in \operatorname{Ran}_{\mathbb{C}} G$. Put $v = G^\dagger \psi$. The exact quaternionic congruence

$$\begin{pmatrix} I_2 & 0 \\ -v^* & 1 \end{pmatrix} \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} \begin{pmatrix} I_2 & -v \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} G & 0 \\ 0 & \lambda - \psi^* G^\dagger \psi \end{pmatrix}$$

proves positivity and rank. By Theorem 2.1, the last scalar is exactly (25). If $G = 0$, then $z = w = 0$, leaving the ray (28) after returning through X. □

Corollary 3.2 (Boundary strata)**.** *The open positive-definite stratum has real dimension* 13*. The generic rank-two boundary stratum is given by $G > 0$ and $s = 0$ and has dimension* 12*. The rank-one locus has top dimension* 7*, obtained from* $\operatorname{rank} G = 1$ *and $s = 0$; the ray* (28) *is contained in its closure.*

4 Lorentz and internal symmetry

For $g \in SL_2(\mathbb{C})$ retain

$$T_P(g) = X(g \oplus 1)X^{-1} = R_P g P + \Pi_u. \tag{29}$$

For $k \in SU(2)$ act on the two-spinor matrix $Z = (z\ w)$ on the right.

Theorem 4.1 (Exact symmetry group)**.** *The action*

$$(g,k)\cdot(G,Z,\lambda)=\big(gGg^*,\ gZk,\ \lambda\big) \tag{30}$$

preserves $\mathcal{C}_{13,P}$ and $\mathcal{N}_{13}$. Its kernel is

$$\{(I_2,I_2),(-I_2,-I_2)\}, \tag{31}$$

so the effective group is

$$\boxed{(SL_2(\mathbb{C})\times SU(2))/\mu_2.} \tag{32}$$

The left factor is exactly the previous marked Lorentz action. The right factor is right multiplication of $\psi\in\mathbb{H}^2$ by unit quaternions and rotates the two complex Weyl summands.

Proof. For $g\in SL_2(\mathbb{C})$,

$$\operatorname{adj}(gGg^*)=(g^*)^{-1}\operatorname{adj}(G)g^{-1},$$

so both spinor quadratic terms and $\det G$ are invariant. For $k\in SU(2)$,

$$\operatorname{tr}\big((Zk)^*\operatorname{adj}(G)(Zk)\big)=\operatorname{tr}\big(k^*Z^*\operatorname{adj}(G)Zk\big)=\operatorname{tr}\big(Z^*\operatorname{adj}(G)Z\big).$$

The positivity criterion (24) has the same invariance. The only elements acting trivially on every G are $g=\pm I_2$, and triviality on every Z then forces $k=g$. The quaternionic realization follows from the usual identification of $SU(2)$ with unit quaternions and the decomposition (1). □

5 The old nine-dimensional object inside the new one

Define

$$\iota_9(G,z,\lambda)=(G,z,0,\lambda), \tag{33}$$

$$r_9(G,z,w,\lambda)=(G,z,\lambda). \tag{34}$$

Theorem 5.1 (Split positive continuation)**.** *The map ι_9 is an injective, determinant-preserving, positivity-preserving linear map*

$$\iota_9:\operatorname{Herm}_3(\mathbb{C})_{\succeq 0}\hookrightarrow\mathcal{C}_{13,P}. \tag{35}$$

The coordinate retraction r_9 maps $\mathcal{C}_{13,P}$ into $\operatorname{Herm}_3(\mathbb{C})_{\succeq 0}$ and satisfies

$$r_9\iota_9=\mathrm{id}\,. \tag{36}$$

Their determinants are related by the exact correction

$$\boxed{\mathcal{N}_9(r_9H)-\mathcal{N}_{13}(H)=w^*\operatorname{adj}(G)w\geq 0\quad(H\in\mathcal{C}_{13,P}).} \tag{37}$$

Both maps are proper on their closed positive cones and therefore extend to the one-point compactifications

$$B^9\xrightarrow{\iota_9^+}B^{13}\xrightarrow{r_9^+}B^9,\qquad r_9^+\iota_9^+=\mathrm{id}\,. \tag{38}$$

The retraction does not restrict to a map from the new rank-defect boundary to the old one: if $G=I_2$, $z=0$, $w\neq 0$, and $\lambda=\|w\|^2$, then $\mathcal{N}_{13}=0$ but $\mathcal{N}_9=\lambda>0$.

Proof. The first statements follow immediately from Theorems 2.2 and 3.1. Since $G\succeq 0$ implies $\operatorname{adj}(G)\succeq 0$, the correction is nonnegative. To prove properness of r_9 on the cone, let (G,z,λ) range over a compact set. If g_i are the eigenvalues of G, (24) gives, in a G-eigenbasis,

$$|w_i|^2\leq\lambda g_i.$$

Thus w is bounded; the fibre is also closed. Properness of ι_9 is immediate. The displayed counterexample proves the final assertion and supplies the exact map relating the two boundaries when a boundary-to-boundary retraction fails. □

5.1 A family of old slices

For a line $L = [\ell] \in \mathbb{CP}^1$, choose $\ell \in \mathbb{C}^2$ with $\ell^*\ell = 1$ and set

$$\mathcal{J}_9(L) = \{(G, Z, \lambda) : Z = \psi\ell^*,\ \psi \in \mathbb{C}^2\}. \tag{39}$$

This subspace is independent of the phase of ℓ.

Proposition 5.2 ($\mathbb{CP}^1$-family of determinant-preserving slices)**.** *Each $\mathcal{J}_9(L)$ is a nine-dimensional copy of* $\mathrm{Herm}_3(\mathbb{C})$*, and*

$$\mathcal{N}_{13}(G, \psi\ell^*, \lambda) = \lambda \det G - \psi^* \operatorname{adj}(G)\psi. \tag{40}$$

The internal $SU(2)$ acts transitively on this family, with stabilizer $U(1)$, so its parameter space is $SU(2)/U(1) = \mathbb{CP}^1$. The locus contained in at least one such slice is exactly $\operatorname{rank}_{\mathbb{C}} Z \le 1$.

Proof. One has

$$\operatorname{tr}\big((\psi\ell^*)^* \operatorname{adj}(G)(\psi\ell^*)\big) = \operatorname{tr}\big(\ell\psi^* \operatorname{adj}(G)\psi\ell^*\big) = \psi^* \operatorname{adj}(G)\psi.$$

The remaining statements are the standard transitive action on complex lines and the rank-one factorization of a 2×2 complex matrix. □

6 The rank-defect sphere is exactly S^{12}

Let

$$\partial\mathcal{C}_{13,P} = \{H \in \mathcal{C}_{13,P} : \det_{\mathbb{H}} H = 0\} \tag{41}$$

and let

$$\mathcal{J}^0_{13,P} = \{K \in \mathcal{J}_{13,P} : \operatorname{tr} K = 0\}. \tag{42}$$

This is a real vector space of dimension 12.

Theorem 6.1 (Spectral-floor coordinates)**.** *For quaternionic Hermitian K, let $\lambda_{\min}(K)$ denote its least real eigenvalue. The maps*

$$\Phi_{13}(K) = K - \lambda_{\min}(K) I_3, \qquad K \in \mathcal{J}^0_{13,P}, \tag{43}$$

$$\Psi_{13}(H) = H - \frac{\operatorname{tr} H}{3} I_3, \qquad H \in \partial\mathcal{C}_{13,P}, \tag{44}$$

are inverse positively homogeneous homeomorphisms. More generally,

$$\mathcal{C}_{13,P} \xrightarrow{\cong} \mathcal{J}^0_{13,P} \times [0, \infty), \qquad H \longmapsto \left(H - \frac{\operatorname{tr} H}{3} I_3, \frac{\operatorname{tr} H}{3} + \lambda_{\min}\left(H - \frac{\operatorname{tr} H}{3} I_3\right)\right). \tag{45}$$

Consequently

$$\boxed{\mathcal{C}^+_{13,P} \cong B^{13}, \qquad (\partial\mathcal{C}_{13,P})^+ \cong S^{12}.} \tag{46}$$

Proof. The identity matrix lies in $\mathcal{J}_{13,P}$, so both shifts remain in the slice. If K is trace-free, $K - \lambda_{\min}(K) I_3$ is positive semidefinite and singular. If H is positive semidefinite and singular, then

$$\lambda_{\min}\left(H - \frac{\operatorname{tr} H}{3} I_3\right) = -\frac{\operatorname{tr} H}{3},$$

so the maps are inverse. Continuity follows from continuity of ordered Hermitian eigenvalues. Writing $H = K + rI_3$ shows that positivity is equivalent to $r \ge -\lambda_{\min}(K)$; the second coordinate in (45) is exactly the nonnegative excess. The one-point compactifications of $\mathbb{R}^{12} \times [0, \infty)$ and its boundary $\mathbb{R}^{12}$ are B^{13} and S^{12}. □

Theorem 6.2 (The old S^8 and its quaternionic link)**.** *The involution*

$$\sigma(G, z, w, \lambda) = (G, z, -w, \lambda) \tag{47}$$

preserves $\mathcal{N}_{13}$ and $\mathcal{C}_{13,P}$. Its fixed space is $\mathrm{Herm}_3(\mathbb{C})$*, and its fixed set on the compactified rank defect is the previously constructed S^8. Under the spectral-floor coordinates,*

$$\boxed{((\partial\mathcal{C}_{13,P})^+, (\partial\,\mathrm{Herm}_3(\mathbb{C})_{\succeq 0})^+) \cong (S^{12}, S^8)} \tag{48}$$

as the standard unknotted linear pair. Its complement is

$$\boxed{S^{12} \setminus S^8 \cong S^3 \times \mathbb{R}^9.} \tag{49}$$

The S^3 is the unit sphere in the added $w \in \mathbb{C}^2 \cong \mathbb{R}^4$ direction and is the unit-quaternion group $\mathrm{Sp}(1) \cong SU(2)$*.*

Proof. The determinant and positivity formulae are even in w. The fixed locus $w = 0$ is $\mathrm{Herm}_3(\mathbb{C})$ by Theorem 1.1. The trace-free pair is the linear pair

$$(\mathcal{J}^0_{13,P}, \mathrm{Herm}^0_3(\mathbb{C})) \cong (\mathbb{R}^{12}, \mathbb{R}^8).$$

One-point compactification gives the standard pair. Removing the fixed set gives

$$\mathbb{R}^8 \times (\mathbb{R}^4 \setminus \{0\}) \cong \mathbb{R}^8 \times S^3 \times \mathbb{R} \cong S^3 \times \mathbb{R}^9.$$

□

7 It is a generating cubic slice, not a Jordan subalgebra

The ambient quaternionic Euclidean Jordan algebra has the Peirce decomposition

$$\mathrm{Herm}_3(\mathbb{H}) \cong \underbrace{\mathrm{Herm}_2(\mathbb{H})}_{6} \oplus \underbrace{\mathbb{H}^2}_{8} \oplus \underbrace{\mathbb{R}}_{1}. \tag{50}$$

The inclusion $\mathrm{Herm}_2(\mathbb{C}) \subset \mathrm{Herm}_2(\mathbb{H})$ selects the $(1,3)$ Lorentz subspace inside the full $(1,5)$ spin factor and gives exactly $\mathcal{J}_{13,P}$.

Theorem 7.1 (Exact Jordan completion)**.** *The space $\mathcal{J}_{13,P}$ is not closed under the Jordan product $A \circ B = (AB + BA)/2$. In the block basis, take*

$$\psi = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \phi = \begin{pmatrix} 0 \\ \mathbf{j} \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & \psi \\ \psi^* & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & \phi \\ \phi^* & 0 \end{pmatrix}.$$

Then the upper-left block of $A \circ B$ is

$$\frac{1}{2}\begin{pmatrix} 0 & -\mathbf{j} \\ \mathbf{j} & 0 \end{pmatrix}, \tag{51}$$

which is not in $\mathrm{Herm}_2(\mathbb{C})$*. Replacing* $\mathbf{j}$ *by* $\mathbf{k}$ *produces the other missing real direction. Therefore*

$$\boxed{\mathrm{Jord}\langle \mathcal{J}_{13,P} \rangle = \mathrm{Herm}_3(\mathbb{H}).} \tag{52}$$

The exact completion is

$$4 + 4 + 4 + 1 \quad \longrightarrow \quad 6 + 4 + 4 + 1 = 15, \tag{53}$$

where the two added coordinates complete the Lorentz vector from $(1,3)$ *to* $(1,5)$*; the two spinors and scalar are unchanged.*

Proof. The displayed product is a direct multiplication. The slice already contains the full $\mathbb{H}^2$ Peirce half-space, the scalar, and the complex four-dimensional part of $\mathrm{Herm}_2(\mathbb{H})$. The two products with $\mathbf{j}$ and $\mathbf{k}$ generate the two missing off-diagonal components of $\mathrm{Herm}_2(\mathbb{H})$. Hence the Jordan algebra generated is the full ambient algebra. □

8 The other thirteen-dimensional determinant and the exact comparison

The same coordinate vector space has a second natural realization as a complex Hermitian 4×4 slice:

$$\mathcal{K}_{13}(G, Z, \lambda) = \begin{pmatrix} G & Z \\ Z^* & \lambda I_2 \end{pmatrix} \in \mathrm{Herm}_4(\mathbb{C}). \tag{54}$$

It has the same $4+8+1$ coordinate count and the same $SL_2(\mathbb{C}) \times SU(2)$ action, but its determinant is quartic.

Theorem 8.1 (Quartic–cubic comparison)**.** *For all* (G, Z, λ)*,*

$$\det_{\mathbb{C}} \mathcal{K}_{13}(G, Z, \lambda) = \lambda^2 \det G - \lambda \operatorname{tr}(Z^* \operatorname{adj}(G) Z) + |\det Z|^2 \tag{55}$$

$$= \boxed{\lambda \mathcal{N}_{13}(G, Z, \lambda) + |\det Z|^2.} \tag{56}$$

On the old slice $Z = (z\ 0)$*,*

$$\det \mathcal{K}_{13} = \lambda \mathcal{N}_9, \qquad \mathcal{N}_{13} = \mathcal{N}_9. \tag{57}$$

Thus the quaternionic slice is the one that preserves the preceding cubic exactly; the complex rank-four slice adds both a factor λ *and the Pluecker correction* $|\det Z|^2$*.*

The coordinate identity maps the quaternionic positive cone into the complex one:

$$\mathcal{C}_{13,P} \hookrightarrow \{(G, Z, \lambda) : \mathcal{K}_{13}(G, Z, \lambda) \succeq 0\}. \tag{58}$$

It is not onto. For example,

$$G = I_2, \qquad Z = I_2, \qquad \lambda = 1 \tag{59}$$

gives a positive-semidefinite complex 4×4 *block, while the quaternionic condition would require* $1 \geq \operatorname{tr}(I_2) = 2$*.*

Proof. For invertible G or nonzero λ, the block determinant formula gives (55); polynomial continuation gives it everywhere. The last term is $\det(Z^* Z) = |\det Z|^2$. The quaternionic positivity criterion is

$$G \succeq 0, \quad \operatorname{Ran} Z \subset \operatorname{Ran} G, \quad \lambda \geq \operatorname{tr}(Z^* G^\dagger Z).$$

The complex block criterion is

$$G \succeq 0, \quad \operatorname{Ran} Z \subset \operatorname{Ran} G, \quad \lambda I_2 - Z^* G^\dagger Z \succeq 0.$$

Since a positive 2×2 matrix has largest eigenvalue at most its trace, the first scalar inequality implies the second matrix inequality. The example has complex Schur complement zero but violates the scalar trace bound. □

Remark 8.2 (Branch count)*.* On the diagonal locus $Z = 0$ and $G = \mathrm{diag}(x, y)$, the quaternionic cubic is $\lambda x y$, whereas the complex quartic is $\lambda^2 x y$. Thus the thirteen-dimensional quartic model merges two of four diagonal branches into one multiplicity-two branch. Four independently variable simple diagonal branches require a second scalar coordinate, giving at least the fourteen-dimensional block $\mathrm{diag}(x, y, \lambda, \mu)$. This is the exact branch morphism behind the distinction; the equation $z_0 z_1 z_2 z_3 = t$ is not produced merely by writing $4 + 4 + 4 + 1$.

9 Exact relation to the existing colour S^{12}

Let

$$V_{\mathrm{col}} = \mathrm{Mat}_{3\times 2}(\mathbb{C}), \qquad \Gamma(A) = A^*A \in \mathrm{Herm}_2(\mathbb{C})_{\succeq 0}. \tag{60}$$

The left $SU(3)$ Gram quotient is $V_{\mathrm{col}}/SU(3) \cong \mathrm{Herm}_2(\mathbb{C})_{\succeq 0}$. Define the proper map

$$F_{13}(A) = X \begin{pmatrix} \Gamma(A) & 0 \\ 0 & 0 \end{pmatrix} X^* \in \partial\mathcal{C}_{13,P}. \tag{61}$$

Proposition 9.1 (The two S^{12} carriers are connected by a typed map)**.** *The map F_{13} factors exactly as*

$$\mathrm{Mat}_{3\times 2}(\mathbb{C}) \xrightarrow{\Gamma} \mathrm{Herm}_2(\mathbb{C})_{\succeq 0} \xrightarrow{\jmath_P} \partial\mathcal{C}_{13,P}. \tag{62}$$

After one-point compactification it gives

$$\boxed{S^{12}_{\mathrm{col}} \xrightarrow{\Gamma^+} B^4 \xrightarrow{\jmath_P^+} S^{12}_{13}.} \tag{63}$$

This sphere map is based null-homotopic because it factors through the based-contractible ball B^4. Its image is the embedded four-ball face $\jmath_P^+(B^4)$; it is not an identification of the two sphere constructions.

Proof. The factorization is the definition. Properness follows from $\mathrm{tr}\,\Gamma(A) = \|A\|_F^2$ and injectivity/properness of $\jmath_P$ on the cone. The compactified Gram quotient has codomain B^4, and a closed ball contracts to its chosen boundary basepoint while fixing that point. □

Remark 9.2 (Relation to the toroidal map). The toroidal cusp construction already uses the same marked integral matrix P as the angular part of the lattice map $\Phi = P \oplus Q$. The new quaternionic coordinate w has not been assigned an integral period lattice, monodromy, fan ray, or filling homomorphism. Therefore the exact new maps are (7), (35), and (62); a map from the completed toroidal total space to S^{12}_{13} requires additional transition and filling data and is not inferred from the shared symbol P.

10 A precise threshold consequence of the cubic

Let $G \in \mathrm{Herm}_2(\mathbb{C})_{>0}$ and retain a conversion constant $\alpha > 0$. Define

$$p = \alpha G, \qquad m^2 = \det p = \alpha^2 \det G. \tag{64}$$

In Dirac variables put

$$Q_G(\psi_L, \psi_R) = \psi_L^* \,\mathrm{adj}(G)\psi_L + \psi_R^* G \psi_R \geq 0. \tag{65}$$

Then

$$\mathcal{N}_{13} = \lambda \det G - Q_G. \tag{66}$$

Proposition 10.1 (Joint scalar–cubic threshold)**.** *Fix $\lambda_0 > 0$ and $\nu_0 > 0$. On the invariant level set*

$$\lambda = \lambda_0, \qquad \mathcal{N}_{13} = \nu_0, \qquad G > 0, \tag{67}$$

one has the exact lower bound

$$\boxed{m^2 \geq \frac{\alpha^2 \nu_0}{\lambda_0}.} \tag{68}$$

Equality holds exactly when $\psi_L = \psi_R = 0$. At a fixed G above threshold, the spinor fibre is the ellipsoid

$$Q_G(\psi_L, \psi_R) = \lambda_0 \det G - \nu_0, \tag{69}$$

which is diffeomorphic to S^7; it collapses to a point at threshold.

Proof. Equation (66) gives

$$\lambda_0 \det G = \nu_0 + Q_G \geq \nu_0.$$

Since both G and $\mathrm{adj}(G)$ are positive definite, equality holds precisely at the zero Dirac spinor. Above threshold, Q_G is a positive-definite quadratic form on $\mathbb{C}^4 \cong \mathbb{R}^8$, and a positive level is an ellipsoidal S^7. □

Remark 10.2. This is a finite-dimensional invariant threshold, not a quantum-field-theoretic mass gap. A Hamiltonian statement would additionally require one fixed positive Hilbert space, a self-adjoint dynamics, a vacuum projection, and an energy-controlled map from these variables into physical states.

11 Conclusion

The expression $4 + 4 + 4 + 1$ has a rigid continuation of the preceding marked construction. Extend the same observation map P from complex to quaternionic Hermitian matrices, but require its observed block to remain in the original four-dimensional Lorentz space $\mathrm{Herm}_2(\mathbb{C})$. The result is the thirteen-dimensional pullback $\mathcal{J}_{13,P}$, with one Lorentz vector, two Weyl spinors (equivalently one Dirac spinor), and one scalar. Its cubic restricts exactly to the nine-dimensional cubic, its positive cone has compactified rank defect S^{12}, and the old S^8 is the fixed unknotted subspace with quaternionic linking sphere S^3.

The same coordinate count also supports a complex rank-four determinant, but the exact identity

$$\det \mathcal{K}_{13} = \lambda \mathcal{N}_{13} + |\det Z|^2$$

separates the two constructions. The quaternionic slice preserves the cubic ladder; the complex slice carries the quartic and Pluecker correction. Finally, Jordan multiplication generates two further Lorentz-vector coordinates and closes the thirteen-dimensional slice to the full fifteen-dimensional algebra $\mathrm{Herm}_3(\mathbb{H})$. These maps specify what the thirteen-dimensional object is, how it contains the old object, how it connects to the existing colour S^{12}, and exactly where further toroidal or physical data would have to enter.

A Symbolic replay

The accompanying exact-arithmetic script verifies:

- all retained P, R_P, u, X identities;
- the quaternionic splitting $\psi^* A \psi = z^* A z + w^* A w$;
- $\det_{\mathbb{C}} \rho(H) = \mathcal{N}_{13}(H)^2$;
- the Dirac charge-conjugation identity;
- the adjugate covariance and internal $SU(2)$ invariance;

- exact restriction to the old cubic and the determinant correction under r_9;
- the complex rank-four determinant and the Pluecker correction;
- the explicit Jordan nonclosure witness.